

Negative Exponents with Variable Bases

Teacher's Edition

Read This First

Second Thread A day (Math Nation 14.3 and 14.4 under ruling 14): the negative exponent law and multiple laws, with variable bases. The lesson is built on 3.04 — the same table, the same 'reciprocal, not opposite' line — and on yesterday's regrouping of coefficients and variables.

The items are original, aligned to the benchmark text and to the B1G-M for MA.8.AR.1.1: its example $(3x^3y^2)^3 = 27x^9y^6$ is the shape of Notes III's last line and bank 7b; its items $y^3z^4 = 1/(y^3z^4)$ (whiteboard 4) and $(a^3/(4b^7))^5 = a^{15}b^{35}/1,024$ (Example 2 and bank 8c) are used as written; its task 2 (the $x^5 \dots x^{-5}$ table) is Notes I; its task 3 (x^3 vs x^{-3}) is whiteboard 9, as a written contrast.

The leading error today is not the sign error from 3.04 but a new one: moving the coefficient across the bar with the variable ($3x^2 \rightarrow 1/(3x^2)$). Whiteboards 3 and 6 and bank 11 are built to surface it. The sign error is still there (whiteboard 5, option D) and is worth a look.

Every answer is expected 'with positive exponents only'. A student who stops at x^{-5} has applied the law correctly and not finished; the key marks the finished form. Say this once, at Notes III.

Standard and Targets

MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

Benchmark Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

Boundary Integer exponents, negative included; monomial bases; nonzero variables wherever a negative exponent appears. Rational exponents are Algebra 1.

Learning target I can rewrite an algebraic expression with negative exponents using only positive exponents, and apply several laws of exponents to a monomial expression.

Essential question What does a negative exponent on a variable mean, and how do several laws combine in one expression?

Building on MA.8.NSO.1.3 (3.04–3.05) — the negative exponent law with numerical bases; T-A1 — product, power and quotient laws with variable bases.

Working toward 3.08–3.09 — scientific notation, where 10^{-n} carries the small numbers; MA.912.NSO.1.2 — rational exponents in Algebra 1.

Where this sits in the course Second of two MA.8.AR.1.1 days; the benchmark is complete after today. The Grade 8 FAST reports it under Algebraic Reasoning.

Tomorrow scientific notation starts (3.08), where 10^{-n} does the same job for small numbers.

Lesson at a Glance

Windy Hill periods run **53 minutes**. The whiteboard round takes the remainder after the taught blocks — **19 minutes** in this lesson — and IXL takes the last 5. Slide numbers below are read from the deck when this edition is built.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-6	11 min
Example 1	7-8	3 min
Your Turn	9-10	3 min
Example 2	11-12	3 min
Your Turn	13-14	3 min
Whiteboards	15-32	19 min
IXL	33	5 min
Total		53 min

Materials Whiteboards and markers. Calculators are allowed on everything; $4^5 = 1,024$ in Example 2 is the one computation worth a calculator.

Where the lesson lives Notes II — 'only the factor wearing the negative exponent moves'. Whiteboard 6 tells you whether it landed; whiteboard 9 asks for the x^3 / x^{-3} contrast in writing.

No partner or group work anywhere in this lesson. Every Math Nation exploration this lesson draws on was written for pairs; here the same tables are worked from the front and on individual boards.

Vocabulary

negative exponent law — $x^{-n} = \frac{1}{x^n}$ and $\frac{1}{x^{-n}} = x^n$, for nonzero x — the base moves across the fraction bar and the exponent turns positive

simplified form — for this unit: a single monomial, or a single fraction, with every exponent positive and no factor repeated

Warm-Up — spaced retrieval

Four questions, two minutes silent, then the reveal. No reteach longer than one sentence per question.

1. Write as a single monomial: $(3x^2)^3$ **$27x^6$**

retrieves: yesterday — T-A1 — power of a product: the 3 is cubed too

2. Find the value: 5^{-2} **1/25**

retrieves: this week — 3.04 — the negative exponent law with a numerical base: reciprocal, not opposite

3. Which is greater, $\sqrt{80}$ or 9? **9**

retrieves: last unit — Unit 2 — comparing a root to a whole number (2.05): $9^2 = 81 > 80$

4. What is the reciprocal of $\frac{2}{3}$? **$\frac{3}{2}$**

retrieves: prior grade — the prerequisite this lesson needs — Grade 6 — reciprocals (MA.6.NSO.2): the number that multiplies to give 1

Slide-by-Slide Teaching Notes

Slide 1. Negative Exponents with Variable Bases 1 min

- Post the learning target. Say the 'today' line and nothing else yet.

Slide 2. Warm-Up — On your own. Four minutes. No notes. 4 min

- Q2 and Q4 together are the lesson: a negative exponent asks for a reciprocal. Q2 is the numerical version from 3.04; today the same law is applied to x. A student with -25 on Q2 will write $-x^2$ today, and that is the board to find early.

Slide 3. Warm-Up — Answers. 1 min

- Reveal. Ask for the band each question came from only if time allows.

Slide 4. Notes — Copy the table. 4 min

- The 3.04 pattern table again, with x in place of 2: each entry is the previous one divided by x. After $x^0 = 1$ the next division gives $1/x$, then $1/x^2$, then $1/x^3$. Students fill the last three cells before the rule is written. The state guide asks for this table (its task 2: x^5 down to x^{-5} , 'why is $x^0 = 1$ ') and for a written contrast of x^3 and x^{-3} — that contrast is whiteboard 9.
- **OFF THE SLIDE, YOURS TO SAY:** ' x^{-3} is one over x^3 . It is a reciprocal. It is not negative, and it is not $-3x$.'

Slide 5. Notes 4 min

- Each factor with a negative exponent moves across the bar on its own and its exponent turns positive. The coefficient does NOT move unless it has the negative exponent: $3x^{-2}$ is 3 over x^2 , not 1 over $3x^2$. That is today's leading misconception and it is on whiteboard 3 and 6. The guide's item $y^{-3}z^4 = 1/(y^3z^4)$ is the second line.
- **OFF THE SLIDE, YOURS TO SAY:** 'Only the factor wearing the negative exponent moves. Everything else stays put.'

Slide 6. Notes 3 min

- Product and quotient laws are unchanged; the exponents are integers now, so the sums and differences can land below zero. The last step is always to write the answer with positive exponents. $x^4 \cdot x^{-6}$ is x^{-2} , which is $1/x^2$. m^3/m^7 is m^{-4} , which is $1/m^4$ — and ask the class to check it by cancelling three m's from seven.

Slide 7. Example 1 — Multiplying monomials with negative exponents.

1 min

- SPOKEN ONLY. Same regrouping as yesterday: coefficients, x's, y's. Ask what $-3 + 5$ is and what $2 + (-4)$ is before anyone writes — the whole example is integer addition.

Slide 8. Example 1 — Worked. *2 min*

- Add the exponents of each variable, then move the y^{-2} down. A board with $12x^2y^{-2}$ stopped one step early; a board with $12x^{-8}y^6$ subtracted where it should have added; a board with $1/(12x^2y^2)$ moved the coefficient too.

Slide 9. Your Turn — Same steps, your numbers. Boards up when done.

2 min

- Catches the coefficient-mover ($1/(4a^3b^2)$) and the sign-dropper ($4a^7b^4$). $4a^3/b^2$ is right.

Slide 10. Your Turn — Answer. *1 min*

- Reveal; name what a wrong board most likely did (see the note above).

Slide 11. Example 2 — Several laws in one expression — the state guide's item. *1 min*

- SPOKEN ONLY. Three laws: power of a quotient, power of a product (the 4 is raised), and the negative exponent. Ask which to do first (the outside exponent) and what happens to the 4.

Slide 12. Example 2 — Worked. *2 min*

- Raise every factor to the 5th: a^{15} on top, $4^5 = 1,024$ and b^{-35} on the bottom. Then b^{-35} moves up. The B1G-M's own description of the result: 'only two exponents, no negatives'. A board with 20 instead of 1,024 multiplied 4 by 5; a board with $a^{15}/(1,024b^{35})$ never moved the b.

Slide 13. Your Turn — Same steps, your numbers. Boards up when done. *2 min*

- Catches the un-raised coefficient ($x^6y^6/3$) and the b-that-did-not-move ($x^6/(27y^6)$). $x^6y^6/27$ is right.

Slide 14. Your Turn — Answer. *1 min*

- Reveal; name what a wrong board most likely did (see the note above).

Slide 15. Whiteboards · Question 1 of 9 *19 min for the round*

- The law, plain, with a letter. A board with $-x^4$ made the exponent's sign the sign of the expression; a board with $-4x$ multiplied.

Slide 16. Whiteboards · Question 1 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 17. Whiteboards · Question 2 of 9

- The second form. A board with $1/y^5$ made the exponent positive without moving the power.

Slide 18. Whiteboards · Question 2 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 19. Whiteboards · Question 3 of 9

- The coefficient stays. A board with $1/(3x^2)$ moved the 3 as well — today's leading error, and the one to name at the board.

Slide 20. Whiteboards · Question 3 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 21. Whiteboards · Question 4 of 9

- The state guide's own item. A board with $1/(yz)^7$ or $1/(yz^7)$ combined unlike bases.

Slide 22. Whiteboards · Question 4 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 23. Whiteboards · Question 5 of 9

- Diagnostic. D is the board that says Notes I did not land — go back to the table and divide 1 by x twice, out loud.

Slide 24. Whiteboards · Question 5 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 25. Whiteboards · Question 6 of 9

- Diagnostic for two laws at once. B and D are the two coefficient errors; C is the exponent error.

Slide 26. Whiteboards · Question 6 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 27. Whiteboards · Question 7 of 9

- Subtraction landing below zero. A board with m^4 subtracted the wrong way; a board with m^4 stopped one step early. Cancelling three m's from seven confirms the answer.

Slide 28. Whiteboards · Question 7 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 29. Whiteboards · Question 8 of 9

- Working backwards across zero. A board with -3 added 2 and -5 ; a board with -7 has the sign wrong. $-5 + 7 = 2$.

Slide 30. Whiteboards · Question 8 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 31. Whiteboards · Question 9 of 9

- Written. The state guide's new task for this benchmark, done as a written contrast. Full credit: the word reciprocal (or '1 over x^3 ') and both values. A student who

writes -8 for x^{-3} has the day's misconception; one who writes 'opposites' has the 3.04 confusion of opposite with reciprocal.

Slide 32. Whiteboards · Question 9 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 33. IXL — Last five minutes. 5 min

- IXL: whatever is not finished is tonight's practice; SmartScore 67.

The Whiteboard Round — what each board tells you

Nine questions, one at a time, each with its reveal. Boards down until the cue, all up together, scan the back row first, question the blank boards before the wrong ones. Question 9 is written work. Every wrong option on a multiple-choice question is a named error; the benchmark it comes from is cited.

#	Question	Answer	What a wrong board says
1	x^{-4}	$\frac{1}{x^4}$	$-x^4$ — negative exponent read as a negative expression [B1G-M 8.AR.1.1 misconception: negative exponents]; $-4x$ — exponent multiplied into the base
2	$\frac{1}{y^{-5}}$	y^5	$1/y^5$ — the sign was changed but the power was not moved [8.AR.1.1: law applied in one direction only]
3	$3x^{-2}$	$\frac{3}{x^2}$	$1/(3x^2)$ — the coefficient was moved with the variable [B1G-M 8.AR.1.1: negative exponent applies only to its own base]; $-3x^2$ — read as a negative
4	$y^{-3}z^{-4}$	$\frac{1}{y^3z^4}$	$1/(yz)^7$ — exponents on different bases were added [8.AR.1.1: like bases only]; $-y^3z^4$ — read as a negative
5	Which expression is equivalent?	$A - 1/x^2$	(B) $4 - (-6)$ — subtracted instead of adding a negative [product law: add the exponents]; (C) $4 \cdot (-6)$ — multiplied the exponents [power of a power confused with a product]; (D) x^{-2} read as the opposite of x^2 [B1G-M 8.AR.1.1 misconception: negative exponents]
6	Which expression is equivalent?	$A - 4/x^6$	(B) the coefficient was not squared [8.AR.1.1: power of a product]; (C) $-3 + 2$ instead of $-3 \cdot 2$ [power of a power confused with a product of powers]; (D) the coefficient was moved below the bar with the x [B1G-M 8.AR.1.1: negative exponent applies only to its own base]

#	Question	Answer	What a wrong board says
7	$\frac{m^3}{m^7}$	$\frac{1}{m^4}$	$m^4 - 7 - 3$ instead of $3 - 7$ [quotient law: top minus bottom]; m^4 — correct but not in the positive-exponent form asked for
8	Find the value of n : $x^{-5} \cdot x^n = x^2$	$n = 7$	-3 — added the given exponents [8.AR.1.1: product law, solve $-5 + n = 2$]; -7 — sign error in the integer subtraction
9	How are x^3 and x^{-3} related? Explain, and give the value of each when $x = 2$.	x^{-3} is the reciprocal of x^3. When $x = 2$: $x^3 = 8$ and $x^{-3} = 1/8$.	' $x^{-3} = -8$ ' or 'they are opposites' — negative exponent read as a negative [B1G-M 8.AR.1.1 misconception: negative exponents]; ' -6 ' — exponent multiplied into the base

Question Bank — variation and answers

Variation. Questions 1–6 are one move each: a plain negative exponent, one in a denominator, a coefficient that stays, two variables with one negative, a variable in the denominator moving up, three factors moving in two directions. 7–8 add a law before the rewrite (product, power of a product, product with two variables; quotient below zero, quotient with a coefficient, the guide's three-law item). 9 works backwards through zero in each law; 10 evaluates x^2 , x^{-2} and $-x^2$ at a number so that reciprocal and opposite are seen as values. 11 is the diagnostic multiple choice on the coefficient; 12 asks equivalence to $1/x^6$ six ways, with the two wrong options wrong for different reasons. The Additional sheet mirrors each position.

The bank and its Additional sheet are posted, never assigned; they feed the quizzes. Answers below are the same values the keys print — both come from one source.

Question Bank

1. $\frac{1}{a^6}$

2. k^3

3. $\frac{7}{y^4}$

4. $\frac{y^5}{x^2}$

5. m^4n^3

6. $\frac{2qr^2}{p^5}$

7. (a) $x^{-5} = \frac{1}{x^5}$ (b) $27y^{-6} = \frac{27}{y^6}$ (c) $10a^4b^{-4} = \frac{10a^4}{b^4}$

8. (a) $m^{-7} = \frac{1}{m^7}$ (b) $3x^{-2}y^{-3} = \frac{3}{x^2y^3}$ (c) $\frac{a^{15}}{4^5b^{-35}} = \frac{a^{15}b^{35}}{1,024}$

9. (a) $n = 7$ (b) $n = -2$ (c) $n = -4$

10. (a) 9 and $1/9$ (b) -9 . No — x^{-2} is the reciprocal of x^2 , which is $1/9$; $-x^2$ is the opposite of x^2 .

11. **A A**

12. **A, B, C, D A, B, C and D**

Question Bank — Additional

1. $\frac{1}{b^8}$

2. m^4

3. $\frac{9}{x^5}$

4. $\frac{a^4}{c^3}$

5. p^2q^6

6. $\frac{4yz^5}{x^3}$

7. (a) $y^{-7} = \frac{1}{y^7}$ (b) $8m^{-12} = \frac{8}{m^{12}}$ (c) $12x^3y^{-4} = \frac{12x^3}{y^4}$

8. (a) $k^{-7} = \frac{1}{k^7}$ (b) $4a^{-4}b^{-4} = \frac{4}{a^4b^4}$ (c) $\frac{x^6}{27y^{-6}} = \frac{x^6y^6}{27}$

9. (a) $n = 7$ (b) $n = 3$ (c) $n = -5$

10. (a) **64 and 1/64** (b) **-64. No** — x^{-3} **is the reciprocal of** x^3 **, which is** **1/64;** $-x^3$ **is the opposite of** x^3 **.**

11. **A A**

12. **A, B, C, D A, B, C and D**

What the audit against Math Nation found

- All 55 items on the bank, additional bank, warm-up, examples and round were re-derived by sympy at build time with the variables declared as positive symbols (mathcheck: 0 findings).
- B1G-M items used as written: $y^{-3}z^4 = 1/(y^3z^4)$ (whiteboard 4); $(a^3/(4b^7))^5 = a^{15}b^{35}/1,024$, checked $4^5 = 1,024$ and $(-7)(5) = -35$ (Example 2, bank 8c).
- The B1G-M example $(3x^3y^{-2})^3 = 27x^9y^{-6}$ is stated in the guide with a negative exponent left in the result; our items always finish with positive exponents, so the shape is used with the final step added.
- No Math Nation Unit 14 pages were available for audit; nothing here is copied from them.

What changed from the book

- Math Nation 14.3 and 14.4 (two book lessons) → one period under ruling 14; the numerical negative-exponent law is a week old and needs restating, not reteaching.
- The guide's partner-discussion task (x^3 vs x^{-3}) → whiteboard 9, written, per the no-partner-work rule.
- Cool-downs → whiteboard round.